

Exercise 5: Wrap-up and Electrostatics 4

IMPORTANT: Note that this is a compulsory exercise!

This exercise is meant as a wrap-up of the topics covered so far in the course. We encourage you to treat it like an exam, i.e. first try to solve it as best as you can before asking for help or looking things up. This way you can get a sense of how much you can manage on your own and which topics you need to look more into. The exercise will be marked the same way as the others, i.e. pass/fail based on honest attempt.

Good luck!

Exercise 1:

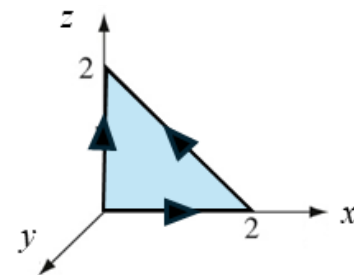
In your own words, define the three fundamental theorems for gradient, divergence and curl. Write down the formulas of the theorems and explain their physical meanings in simple terms.

Exercise 2:

Test Stokes' theorem for the vector field,

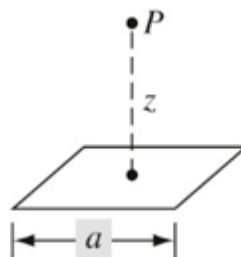
$$\mathbf{v} = xy \hat{\mathbf{x}} + 2yz \hat{\mathbf{y}} + 3zx \hat{\mathbf{z}},$$

using the triangular surface shown in the figure.



Exercise 3:

Find the electric field a distance z above the center of a square loop (side a) carrying a uniform line charge λ .



Exercise 4:

Consider an infinitely long hollow cylindrical tube with radius R , which carries a uniform surface charge σ . Use Gauss's law to find the $\mathbf{E}$ -field inside and outside the tube.

Exercise 5:

Find the electric potential V at a distance z above the center of the charge distributions shown in the figures below. In each case, compute, $\mathbf{E} = -\nabla V$.

The configurations are:

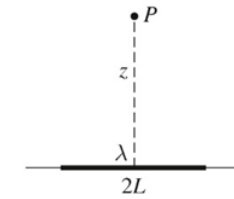
- (a) Uniform line charge λ , length $2L$.

[Hint: $\int \frac{dx}{\sqrt{z^2 + x^2}} = \ln(x + \sqrt{z^2 + x^2})$

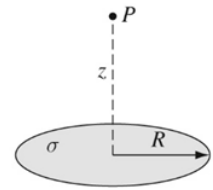
and $\frac{d}{dz} \ln\left(\frac{A}{B}\right) = \frac{A'}{A} - \frac{B'}{B}$]

- (b) Uniform surface charge σ , disk of radius R .

[Hint: $\int \frac{r dr}{\sqrt{z^2 + r^2}} = \sqrt{z^2 + r^2}$]



(a) Uniform line charge



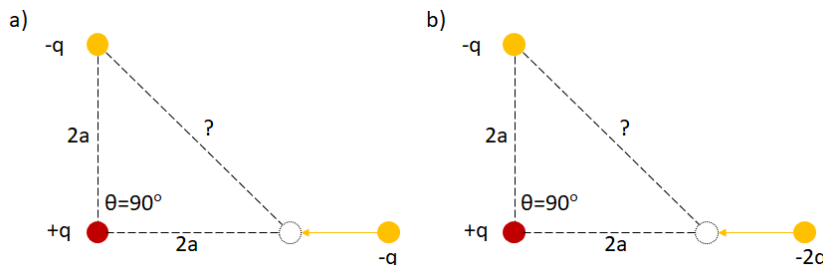
(b) Uniform surface charge

Exercise 6:

Two opposite charges (point charges) are spaced by $2a$.

- a) i) What is the work (W) that is required to bring another charge of $-q$ to form an orthogonal triangle as shown in the figure below. ii) What is the work required to assemble the whole configuration of charges?
b) Same as above. Now consider a new charge of $-2q$.

Hint: Find first V . Consider the interaction of all charges with each other.

**Exercise 7:**

Consider two concentric spherical shells of radii a and b . Suppose the inner one carries a charge q , and the outer one a charge $-q$ (both of them uniformly distributed over the surface).

- a) Calculate the energy by interpreting it as the energy stored in the electric field throughout space and thus use

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau,$$

integrating the square of the total field from both shells over all space.

- b) Calculate again the energy using the above formula, but now consider the electric fields of each shell separately. We aim to see the impact of the fact that electrostatic energy does not obey the superposition principle. Find the energy for each shell and calculate the energy for the cross-

term arising when you square the total field explicitly. Show that the result matches that in a) only when you include the cross-term.

Exercise 8:

In lab 1, many of you have used (or will use) a coaxial cable, consisting of an inner and an outer conductor separated by a dielectric as shown to the left in the figure below. Consider the simplified case to the right in the figure, where the medium between the conductors is air. The inner conductor is a solid cylindrical conductor with radius a , and the outer conductor is a thin cylindrical tube with radius b . The cable is infinitely long and has charge Q on the inner conductor within a cable segment of length L .

- Find the electric field in between the conductors.
- Find the potential difference between the inner and outer conductors.
- Show that the capacitance per unit length of the cable is $\frac{2\pi\epsilon_0}{\ln(\frac{b}{a})}$.
- The capacitance per unit length is determined to be 20 pF/m. What is the ratio between the radii of the inner and outer conductors?

